Bounding the Quadtree Size

Why 2m and 2n are upper bounds on the padded dimensions — not padded dimensions themselves

> **KEY POINT**
> 2*m* and 2*n* do **not** need to be powers of two. They are simply upper bounds on the powers of two used for padding.

1. Proof that M′ ≤ 2*m*

Let

$$\mathrm{M}' = 2^{\lceil \log_2 \mathrm{m} \rceil},$$

where $m \geq 1$ is an integer.

By the definition of the ceiling function,

$$\lceil \log_2 \mathrm{m} \rceil < \log_2 \mathrm{m} + 1.$$

Therefore, exponentiating both sides with base 2 (a strictly increasing function),

$$2^{\lceil \log_2 \mathrm{m} \rceil} < 2^{\log_2 \mathrm{m} + 1}.$$

Using $2^{\log_2 m} = m$,

$$\mathrm{M}' = 2^{\lceil \log_2 \mathrm{m} \rceil} < 2m.$$

Hence, in fact,

$$\mathrm{M}' < 2m$$

for every positive integer m. Therefore the weaker statement

$$\mathrm{M}' \leq 2m$$

is certainly true.

The same argument gives

$$\mathrm{N}' < 2n.$$

2. Why This Makes Intuitive Sense

Suppose 2^k is the smallest power of two that is at least m. Then

$$2^{k-1} < m \leq 2^{k}.$$

Multiplying the left inequality by 2,

$$2^{k} < 2m.$$

Since $M' = 2^{k}$,

$$M' < 2m.$$

Examples

m	M'	$2m$	Inequality
4	4	8	$4 < 8$
5	8	10	$8 < 10$
10	16	20	$16 < 20$
17	32	34	$32 < 34$

So $2m$ is not itself a padded dimension. It is merely a convenient upper bound.

3. Applying This to the Quadtree

The padded dimensions satisfy

$$M' < 2m, \quad N' < 2n.$$

Therefore the number of padded cells (and hence the maximum possible number of leaf cells, if every cell becomes a leaf) is

$$L = M' N'.$$

Thus

$$L < (2m)(2n) = 4mn.$$

So

$$L < 4mn.$$

If you only need a non-strict bound, you can write

$$L \leq 4mn.$$

> **ONE IMPORTANT CORRECTION TO THE NEXT STEP**

> Your subsequent argument
>
> $$\mathrm{T} = 1 + 4 + 4^2 + \ldots + 4^{\mathrm{H}} = (4^{\mathrm{H}+1} - 1)/3 \approx (4/3)\mathrm{L}$$
>
> is correct **only** when the quadtree is a complete/full 4-ary tree with all leaves at the same depth.

4. A Cleaner Bound via Full 4-ary Trees

There is an even cleaner way to obtain the bound. For a full 4-ary tree with L leaves, let I be the number of internal nodes. Since every internal node has exactly four children,

$$4I = I + L - 1,$$

because a tree with $I + L$ total nodes has $I + L - 1$ edges. Hence

$$3I = L - 1,$$

so

$$I = (\mathrm{L} - 1)/3.$$

Therefore the total number of nodes is

$$\mathrm{T} = \mathrm{I} + \mathrm{L} = (\mathrm{L} - 1)/3 + \mathrm{L} = (4\mathrm{L} - 1)/3.$$

Since $L < 4mn$,

$$\mathrm{T} = (4\mathrm{L} - 1)/3 < (16mn - 1)/3 < (16/3)\,mn.$$

Thus a convenient asymptotic bound is

$$\mathrm{T} = \mathrm{O}(\mathrm{mn}).$$

5. Summary of the Overall Reasoning

$$\mathrm{M}' < 2m, \quad \mathrm{N}' < 2n \longrightarrow \mathrm{L} = \mathrm{M}'\mathrm{N}' < 4mn \longrightarrow \mathrm{T} = (4\mathrm{L} - 1)/3 < (16/3)\,mn$$

> **THE CRUCIAL IDEA**
>
> $2m$ and $2n$ are **bounds**, not powers of two themselves.